

2023 AMC 10A Problem 17 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 17. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 17

Problem:

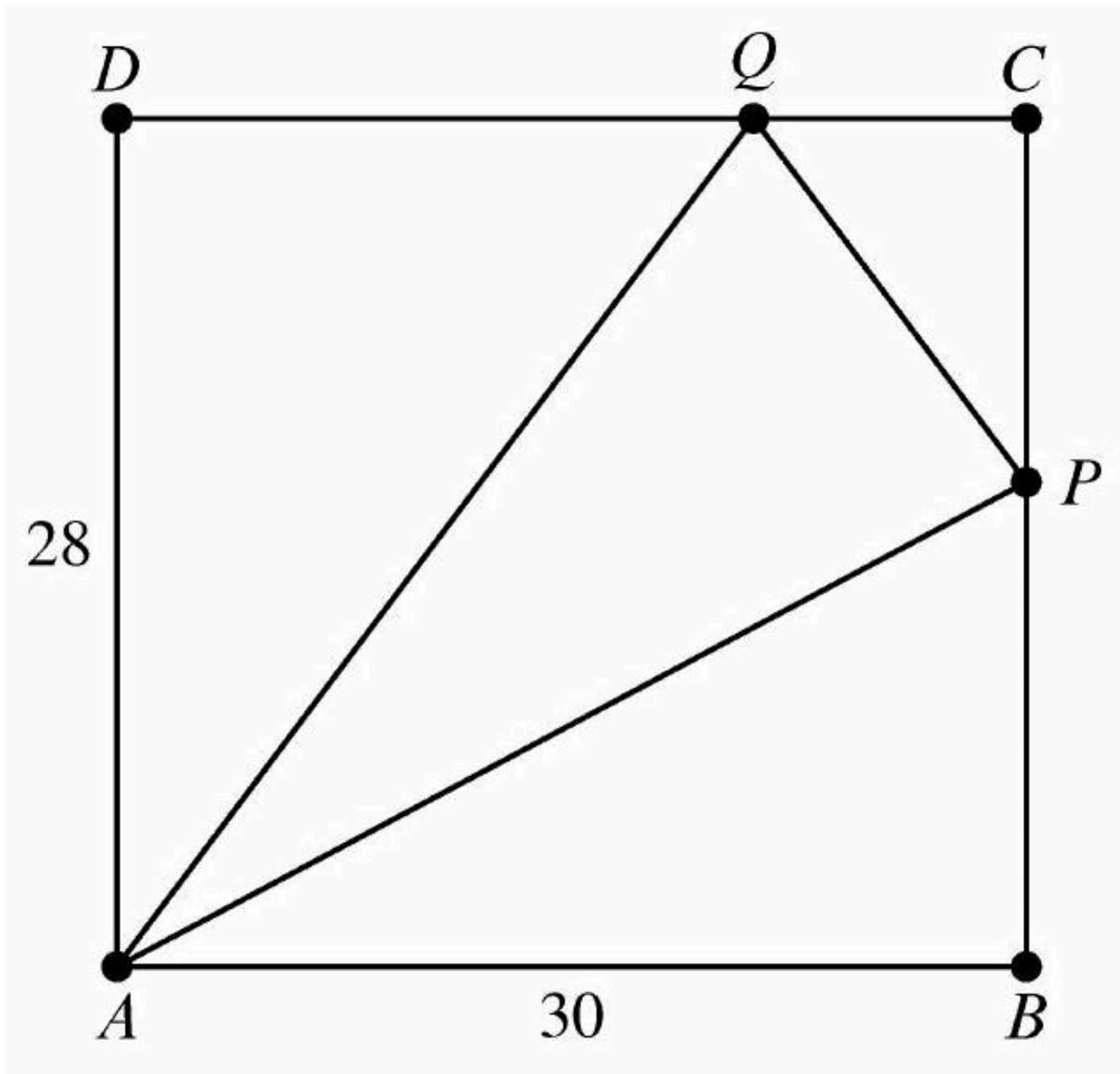
Let $ABCD$ be a rectangle with $AB = 30$ and $BC = 28$. Points P and Q lie on $\overline{BC}$ and $\overline{CD}$, respectively, so that all sides of $\triangle ABP$, $\triangle PCQ$, and $\triangle QDA$ have integer lengths. What is the perimeter of $\triangle APQ$?

Answer Choices:

- A. 84
- B. 86
- C. 88
- D. 90
- E. 92

Solution:

The figure below shows the given information.



The Pythagorean triples with smallest total lengths are $3 - 4 - 5$, $5 - 12 - 13$, $7 - 24 - 25$, $8 - 15 - 17$, and multiples thereof. Note that $AD = 28 = 4 \cdot 7$. Then triangle $\triangle ADQ$ must be a multiple of $3 - 4 - 5$ (with multiplier 7) or a multiple of $7 - 24 - 25$ (with multiplier 4). But the latter

option implies $DQ = 4 \cdot 24 > 30$, so the first option is correct. This gives $AQ = 5 \cdot 7 = 35$.

Because $DQ = 3 \cdot 7 = 21$, it follows that $QC = 30 - 21 = 9$. Thus triangle $\triangle QCP$ is also a multiple of $3 - 4 - 5$ (with multiplier 3). This gives $PC = 4 \cdot 3 = 12$ and $PQ = 5 \cdot 3 = 15$.

Finally, because $PC = 12$, it follows that $PB = BC - PC = 28 - 12 = 16$. Then triangle $\triangle ABQ$ must be a multiple of $8 - 15 - 17$ (with multiplier 2), so $AP = 17 \cdot 2 = 34$. The perimeter of $\triangle APQ$ is $34 + 15 + 35 = (A) \boxed{84}$.

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